



SINA OJAGHI *Sabbatical PhD Students*

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📍 P23, marvi, W kamali, Pounak, Tehran, Iran 📅 1995-09-19 🧑 Single
🛡 Educational Exemption 🔗 sinaoj.com 🔗 linkedin.com/in/sina-ojaghi
🐙 github.com/sinaojaghi 🏠 Sina Ojaghi

📄 Summary

As a Ph.D. Candidate in Computer Architecture at the University of Tehran, my academic journey has been deeply rooted in research and education. Having served as an instructor at both Hamedan University of Technology and the University of Tehran, I have contributed to my students' academic growth while continuously developing my teaching and research skills. Alongside academia, I currently work as an IT Manager at HamVta Arya Co. and as a Hardware Programmer at Afra Turbine Energy Co., applying my technical expertise to real-world challenges. Driven by a passion for discovery and innovation, I am actively engaged in research and development, with a particular focus on hardware security and spiking neural networks. I seek to further expand my research capabilities and contribute meaningful insights to these emerging fields.

🎓 Education

PhD in Computer Engineering,
Tehran University
2020 – Present | Tehran, Iran
Branch: Computer Architecture GPA: 18.05

Master of Computer Engineering, Islamic Azad University Science and Research Branch
2017 – 2019 | Tehran, Iran
Branch: Computer Architecture GPA: 18.45

Bachelor of Computer Engineering,
Hamadan University of Technology
2013 – 2017 | Hamadan, Iran
Branch: Computer Architecture GPA: 16.34

🌐 Language

English — Score 513 on the Tolimo language test (2023) Reading Writing Speaking Listening • **Persian** — Native speaker Reading Writing Speaking Listening

📁 Work Experience Academic

Instructor Network Laboratory, University of Tehran
09/2024 – Present | Tehran, Iran

Computer Networks Laboratory Instructor at the University of Tehran, teaching undergraduate computer networks laboratory courses and supervising hands-on networking experiments.

Research Assistant, Institute for Research in Fundamental Sciences (IPM)
09/2024 – Present | Tehran, Iran

Research Assistant at the Institute for Research in Fundamental Sciences (IPM), conducting research under the supervision of Dr. Ahmad Khonsari and Dr. Kamyar Givaki.

Research Assistant, University of Tehran
04/2024 – Present | Tehran, Iran

Research Assistant at the University of Tehran, focusing on FPGA hardware security, side-channel attacks, and PUFs under Dr. Siamak Mohammadi.

Instructor Artificial Intelligence (AI), Hamedan University of Technology
09/2021 – 10/2023 | Hamadan, Iran

Department of Computer and IT Engineering and Department of Electrical Control and Electronics Engineering

Instructor Research Methods (RM), Hamedan University of Technology
09/2021 – 10/2023 | Hamadan, Iran

Department of Electrical Control and Electronics Engineering and Department of Computer and IT Engineering

Secretary and speaker of Cyber-Physical Systems society if iran, CPSSI
03/2022 – Present | Tehran, Iran

link: <https://cpssi.ir/>

Idea Developer and Senior Consultant of Targeted Internship Site,
Hamedan University of Technology

09/2016 – 01/2019 | Hamadan, Iran
link: <http://intern.hut.ac.ir/>

📁 Research and Publications

Design and Implementation of Remote Side-Channel Attacks and Countermeasures on Spiking Neural Networks, Ph.D. Proposal
Publisher: University of Tehran

Design and manufacture of audio processors for smartening domestically produced vehicles, Master Thesis
Publisher: Islamic Azad University, Science and Research Branch

Honors

13th place among the candidates in the 2020 PhD entrance exam for Computer Architecture
2020

1st place in the M.S. class of 2017, Computer Architecture, Islamic Azad University, Science and Research Branch
2017

2nd place in the B.S. class of 2013, Computer Architecture, Hamedan University of Technology
2017

1st place in the ACM Competitions, Hamedan province
2015

Certificates

The Top project of the computer department of Hamadan University of Technology to enter the Elcamp exhibition in Hamadan province
2016 Institute: Hamadan University of Technology

The Top project of the computer department of Hamadan University of Technology to enter the exhibition of research, technology and market technology of Hamadan province
2016 Institute: Hamadan University of Technology

The Top project of the Computer Department of Hamadan University of Technology Smartening of smart hospitals to lead to a conference on the role of economic prosperity in the prevention of social crimes
2017 Institute: Hamadan University of Technology

Transfer voice data under secure protocol and use it in smart homes, Bachelor Thesis
Publisher: Hamedan university of technology

Evaluating the Impact of Active Wire Fence Placement on Remote Power Side-Channel Attacks in Multi-Tenant FPGAs
Ojaghi, Sina, et al. 2026 Fourth International Conference on Distributed Computing and High Performance Computing (DCHPC). IEEE, 2026. Link: <https://ieeexplore.ieee.org/abstract/document/11517259/>

Object Detection for Vehicles with Yolo
Maleki, Pouria, Abbas Ramazani, Hassan Khotanlou, Sina Ojaghi, Milad Mousavi, Alexey Kalinin, and Amir Mosavi. 2024 IEEE 22nd World Symposium on Applied Machine Intelligence and Informatics (SAMI): IEEE, 2024. Link: <https://ieeexplore.ieee.org/document/10432884>

Iranian Vehicle Images Dataset for Object Detection Algorithm
Maleki, Pouria, Abbas Ramazani, Hassan Khotanlou, and Sina Ojaghi. Journal of AI and Data Mining 12, no. 1 (2024): 127-36. Link: https://jad.shahroodut.ac.ir/article_3164.html

Stress-Strain Modeling of an Al-Mg Using an Artificial Neural Network
Ojaghi, Sina, Farzan Barati, and Somaye Jafari jassbi. Paper presented at the 3rd International Conference on Electrical, Computer and Mechanical Engineering papers, 2020. Link: <https://civilica.com/doc/1128854/>

Modeling of Az52 Flow Curve at High Temperature by Implicating the Genetic Algorithm Method
Ojaghi, Sina, Farzan Barati, and Somaye Jafari jassbi. Paper presented at the 3rd International Conference on Electrical, Computer and Mechanical Engineering papers, 2020. Link: <https://civilica.com/doc/1128855/>

Offline voice detection in smart homes
Ojaghi, Sina, and Javid Ghafourian. The 4th International Conference on Electrical Engineering, Computer, Mechanics and Artificial Intelligence. 2023. Link: <https://en.civilica.com/doc/1780938/>

Teaching Experience in Universities, Schools, and Academic Associations

Teaching Assistant for Master's and PhD Programs at University of Tehran

- Hardware Security and Trust
Fall 2022, Fall 2023
- Functional Verification of HDL Models
Fall 2021, Fall 2022
- Performance Evaluation
Spring 2022, Spring 2023, Spring 2024, Spring 2025
- Computer Networks (B.Sc.)
Fall 2021, Fall 2022, Fall 2023, Fall 2024, Fall 2025

Teaching Assistant for Bachelor Programs at Hamedan University of Technology

- Microprocessor and Assembly Language
September 2015 - June 2016
- Computer Architecture
December 2015 - December 2016
- Logic Circuits
December 2015 - June 2016
- Discrete Mathematics
December 2015 - June 2016

Skills

VHDL Programming Language

Expert

Matlab Programming Language

Expert

Python Programming

Advanced

Embedded linux Raspberry Pi

Advanced

System Verilog

Advanced

Arm Programming

Advanced

Altium Designer

Intermediate

Programming Language C++, C, Java

Intermediate

Cisco CCNA

Intermediate

Teaching for Association of Computer Science at Hamedan University of Technology

- Data Transfer and Telecommunications

December 2016 - June 2017

- Microprocessor and Assembly Language

September 2016 - June 2017

- Computer Architecture

February 2015 - January 2017

- Digital Electronics

December 2016 - June 2017

- Logic Circuits

September 2014 - December 2015

- Algorithm Design

December 2015 - June 2016

- C Programming Language

September 2014 - December 2015

Teacher

- IOT

School Khadije Kobra Tehran Tehran, Iran September 2017 - May 2019

- Computer

School Dar ol Fonoon Hamedan Hamedan, Iran September 2014 - May 2016

Work Experience Industry

Hardware Programmer

09/2022 – Present | Hamedan, Iran

Afra Turbine Energy Co.

- Designed and implemented embedded software for microcontroller-based hardware systems

- Developed firmware updates to improve the functionality and performance of hardware devices

IT Manager

09/2023 – 06/2024 | Tehran, Iran

HamVeta Co.

- Developing and implementing IT policies and procedures to streamline processes and enhance security

R & D

11/2019 – 04/2020 | Tehran, Iran

Nabz

- link: <https://nabzgroup.com/>

Idea Developer and Senior Consultant of Targeted Internship Site

09/2016 – 01/2019 | Hamedan, Iran

Hamedan University of Technology

- link: <http://intern.hut.ac.ir/>

Supporter, Group Manager

09/2015 – 05/2017 | Hamadan, Iran

Qalam Chi Educational Foundation

Laptop and Computer Software and Hardware Repairs

05/2013 – 09/2014 | Hamadan, Iran

DPC

Dr. Siamak Mohammadi,
Computer department group manager
Tehran University
smohammadi@ut.ac.ir

Dr. Ahmad Khonsari,
Member of the Scientific Council of the IPM
Basic Knowledge Research Institute
ak@ipm.ir

Dr. Somaye Jafar Ali Jasbi,
Computer department group manager
Islamic Azad university science and
research branch
s.jassbi@srbiau.ac.ir

Dr. Ahmad Khadem Zade,
Deputy of Iran Telecommunication Research
Center and member of the Scientific Council
of Islamic Azad University, Science and
Research Branch
a_khademzadeh@azad.ac.ir

Dr. Ehsan Aerabi,
PhD student of Grenoble University of
Technology and former faculty of Hamadan
University of Technology
e_aerabi@comp.iust.ac.ir

Dr. Mir Hossin Dezfoulial,
Computer department group manager Bu
ali sina university
dezfoulial@basu.ac.ir

Training and replacing the heavy part of the circuit with a neural network and accelerating the circuit

06/2021

For: Under Supervision: Dr. Mehdi Modaresi (University of Tehran) Chip multiprocessor (CMP) Advanced lesson project

Investigation and detection of hardware malware in AES-T300 RS232-T400

06/2021

For: Under Supervision: Dr. Siamak Mohammadi (University of Tehran) Hardware Security and Trust Advanced lesson project

Verification Calculator II & III & RISC-V

01/2021

For: Under Supervision: Dr. Siamak Mohammadi (University of Tehran) This verification includes: circuit performance simulation, design of a Stimuli production platform, Golden Model design, Checker design, Scoreboard design and implementation of Coverage techniques

Automatic annotation of photos with genetic algorithm in MATLAB environment

2017

For: Under Supervision: Dr. Mohsen Fathiyan (Hamadan University of Technology) computer vision Advanced lesson project and this project was used to identify the status of the signature and the owner of the signature in banking operations.

Ideation, Design and Implementation of Targeted Internship Site

2016

For: Hamadan University of Technology The site we designed worked for about 3 years at the university and then became inaccessible due to the more comprehensive Golestan system. Link: <http://intern.hut.ac.ir>

Data transfer in FPGAs (UART) in VHDL language in ISE environment

2015

For: Under Supervision: Dr. Mohsen Ahmad Vand (Hamadan University of Technology) Data transfer Advanced lesson project

MIPS processor design with some commands in VHDL language in ISE environment

2009

For: Under Supervision: Dr. Ehsan Aerabi (Hamadan University of Technology) Computer architecture Advanced lesson project